

HARJEET SINGH PANNU

ELECTRONICS ENGINEER • VLSI & EMBEDDED SYSTEMS • HARDWARE DESIGN ENGINEER

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PROFESSIONAL SUMMARY

B.Tech graduate in Electronics & Communication Engineering from VIT Chennai (CGPA 9.5+/10), certified in VLSI Design & Verification, with hands-on project experience across RF/antenna design, embedded systems, IoT, and AI-integrated hardware. Ranked #79 of 76,886 participants worldwide in the WorldQuant Quantitative Championship 2025, reflecting strong analytical and problem-solving ability. Completed a 6-month Research Analyst internship at JPMorgan Chase, building rigor in structured analysis and technical communication. Seeking an entry-level Hardware / Semiconductor Engineering role to apply core electronics, VLSI, and embedded systems expertise.

PROFESSIONAL EXPERIENCE

JPMorgan Chase & Co. — Commercial & Investment Banking Division

Jan 2026 – Jun 2026

CRG Research Analyst Intern

Mumbai, India

- Conducted structured research and data analysis, translating complex technical and business information into clear, actionable insights.
- Authored technical reports and analytical presentations reviewed by senior stakeholders across a global organization.
- Applied quantitative and qualitative frameworks to identify patterns and trends across large, complex datasets.
- Collaborated within fast-paced, cross-functional global teams, adapting quickly to new tools, domains, and problem sets.

EDUCATION

Vellore Institute of Technology (VIT), Chennai

2022 – 2026

B.Tech, Electronics & Communication Engineering

CGPA: 9.5+ / 10.0

- Academic Excellence Award — Top Performer, B.Tech ECE

Apeejay School, Kolkata

2021 – 2022

CBSE Class XII

Percentage: 98%

Podar International School, Aurangabad

2019 – 2020

CBSE Class X

Percentage: 96%

SKILLS & COMPETENCIES

Hardware & VLSI: Verilog, VLSI Design & Verification, Cadence, ModelSim, Circuit Design

RF & Embedded: ANSYS HFSS, Embedded Systems, ESP32, Raspberry Pi, Wireless Communications

Programming: C++, Python, Java, MATLAB

Tools & Platforms: Arduino IDE, Keil uVision, Git, VS Code, Postman

Analytical: Data Analysis, Quantitative Research, Technical Documentation

HARDWARE & TECHNICAL PROJECTS

Stepped Bowtie Antenna for Underwater Wireless Communication

- Designed and simulated a 10 GHz stepped bowtie antenna in ANSYS HFSS, tested across air, saline, salt, and distilled water environments.

IntelliWeed — AI-Powered Autonomous Agricultural Robot [\[Brochure\]](#) [\[Demo\]](#)

- Engineered an embedded computer-vision system using YOLOv8, OpenCV, and Raspberry Pi for autonomous weed detection, achieving 90% detection accuracy and reducing manual labor by 40%.

Smart Plant Health Monitoring System — Industrial IoT Platform

- Developed a real-time IoT sensing and automated irrigation-control platform using Arduino and Blynk, improving water-usage efficiency by 50% and enhancing crop-yield forecasting.

ACHIEVEMENTS & CERTIFICATIONS

- Certified in VLSI Design & Verification (Verilog) — Ethnus, industry-recognized chip design methodology [\[Verify\]](#) (Code: FRFGXK4K)
- WorldQuant Quantitative Championship 2025 — Global Rank #79 of 76,886 participants [\[Award Certificate\]](#) [\[Rank Certificate\]](#)
- Academic Excellence Award — Top Performer, B.Tech ECE, VIT Chennai
- Patent Under Review — Smart Multi-Biometric UPI Authentication System
- International Research Finalist — ICDTSA Conference [\[Certificate\]](#)
- 1st Runner-Up, Pitch-A-Thon — competed against 70+ teams [\[Certificate\]](#)

LEADERSHIP & EXTRACURRICULAR

Sponsorship & Outreach Lead — Fraternity of Young Innovators (FYI)

- Led industry outreach and corporate sponsorships for the club, developing stakeholder management and technical communication skills.